MATIHIKO MAHI LIMITED · HIGHLAND BAGPIPER

Stripe Integration & GST Fee Structure Analysis

A technical and financial analysis of the platform's payment configuration, GST compliance requirements, and recommended fee structure for New Zealand operations.

Prepared for: Matihiko Mahi Limited

Platform: Highland Bagpiper

Date: 24 March 2026

Status: Implementation Planning

NAVIGATION

Contents

1 Context & Questions

2 Current Stripe Configuration

3 GST Roles & Obligations

4 Fee Analysis at $300 Booking

5 Platform Fee Recommendation

6 Implementation Plan

7 IRD Compliance Notes

SECTION 1

Context & Questions

The following questions were raised by the platform owner while reviewing the Highland Bagpiper Stripe integration. This report addresses each in full.

> **QUESTION 1 — STRIPE PAYMENT METHOD**
>
> *On my payment intent, am I using the Stripe account field or the transfer destination field?*

The current implementation uses **transfer_data.destination**. This means the full charge is collected by the Highland Bagpiper platform account first, then Stripe automatically transfers the piper's portion to their connected account. This is distinct from *on_behalf_of* (which charges directly on behalf of the connected account) or *stripe_account* (which makes API calls as the connected account).

Current Configuration

File: `convex/stripe.ts` · Line 98 — `transfer_data: { destination: piper.stripeAccountId }`

The platform collects 100% of the charge, retains the application fee, then transfers the remainder to the piper.

> **QUESTION 2 — GST & FEE STRUCTURE**
>
> *The Piper is the merchant and may be GST registered or not. The client may be GST registered or not. Matihiko Mahi Limited is the platform owner and is GST registered. The payment will go straight from client to the merchant. A platform fee (currently set at 5%) needs to be deducted from the client to pay for the platform, the Stripe fees and GST. We will likely need the piper to indicate if they are GST registered when setting up their profile with GST number and their company name. Can you please create a plan based on the current platform configuration to a working system that makes it really simple and easy for the piper to manage. Based on an average charge of $300 from the piper per event, please analyse if 5% is enough for the platform to make a small profit.*

The analysis below addresses each component of this question: GST obligations for each party, a full fee breakdown at $300, an assessment of the 5% rate, a recommendation, and a step-by-step implementation plan.

SECTION 2

Current Stripe Configuration

Payment Flow Architecture

The Highland Bagpiper platform uses **Stripe Connect with transfer_data.destination**. This architecture is correct and appropriate for a marketplace where the platform takes a fee. The flow works as follows:

Step	What Happens	Who It Involves
1	Customer completes Stripe Checkout	Customer → Platform Stripe account
2	Stripe deducts its processing fee from the total	Stripe → Platform
3	Platform retains the application_fee_amount	Platform keeps its fee
4	Remainder is transferred to the piper's connected account	Platform → Piper Stripe account

Current Fee Settings

Setting	Current Value	Notes
Platform fee rate	5%	Of piper's base fee
GST handling	Not implemented	No GST fields on piper profile
Fee includes GST?	No	Needs to be added — platform is GST registered
Stripe product (NZ domestic)	1.7% + $0.30 NZD	Charged on the total amount
Stripe product (international)	3.5% + $0.30 NZD	Relevant for tourist bookings

Important: Stripe Fees Come Off the Platform's Share

With transfer_data.destination, Stripe's processing fee is charged to the **platform account**, not the piper's connected account. This means Stripe's fee reduces the platform's net income — it must be factored into the platform fee rate calculation.

SECTION 3

GST Roles & Obligations

Party-by-Party GST Summary

Party	GST Status	Obligation
Matihiko Mahi Limited (Platform)	**GST Registered**	Must charge GST on the platform service fee and remit to IRD. Issues GST invoices for the platform fee portion.
Piper (GST Registered)	**GST Registered**	Their performance fee is GST-exclusive. They must collect and remit 15% GST on their fee to IRD independently.
Piper (Not GST Registered)	Not Registered	No GST obligation. Their fee is flat — no GST component. No IRD filing required for GST.
Customer (Either Status)	Either	Pays the GST-inclusive total. If GST registered, they may claim the platform fee GST as an input tax credit (the piper GST portion depends on piper's registration).

What "GST Registered" Means for Pipers

A piper registered for GST in New Zealand is required to register once their annual turnover from taxable supplies exceeds $60,000. Once registered:

- Their performance fee is quoted **excluding GST**
- 15% GST is added on top when the customer pays
- The piper files a GST return with IRD (usually every 2 months) and remits the collected GST
- The piper can also claim back GST on their own business expenses

Platform's Responsibility

The platform should clearly communicate to pipers: "If you are GST registered, the fee you enter is treated as GST-exclusive. We add your GST for you at checkout. You are responsible for filing and remitting that GST with IRD."

Key Design Principle

The platform should keep it simple for pipers: **they enter one number — their fee**. The system handles whether GST is added on top based on their registration status. The piper never needs to manually calculate or enter GST amounts.

SECTION 4

Fee Analysis at $300 Booking

NZ Stripe Processing Fees

Card Type	Rate	On $300 charge	On $315 charge (5% fee)
Domestic NZ card	1.7% + $0.30	$5.40	$5.655
International card	3.5% + $0.30	$10.80	$11.325

Note: Tourist bookings (international cards) may represent 30–40% of bookings given the nature of the service. This materially affects platform profitability and must be considered in fee-setting.

Scenario A — Piper NOT GST Registered, Current 5% Fee

Customer Pays — Non-GST Piper · $300 Base Fee · 5% Platform Rate

Piper's fee (flat, no GST)	$300.00
Platform fee (5% of piper fee, incl. GST)	$15.00
Customer pays	**$315.00**
↳ Stripe domestic fee (1.7% + $0.30)	−$5.655
↳ Stripe international fee (3.5% + $0.30)	−$11.33
↳ Platform retains (application fee)	$15.00
↳ GST platform must remit to IRD (15/115 of $15)	−$1.96
Piper receives	**~$294.00**
Platform net profit (domestic card)	~$7.39
Platform net profit (international card)	~$1.71

Scenario B — Piper GST Registered, Current 5% Fee

Customer Pays — GST-Registered Piper · $300 Base Fee (excl. GST) · 5% Platform Rate

Piper's fee (excl. GST)	$300.00
GST on piper's fee (15%)	$45.00
Platform fee (5% of base, incl. GST)	$15.00 + $2.25 GST = $17.25
Customer pays	**$362.25**
↳ Platform retains (application fee)	$17.25
↳ GST platform remits to IRD	−$2.25
↳ Stripe domestic fee	−$6.46
Piper receives (before their IRD remittance)	**~$338.00**
Piper's true net (after paying $45 GST to IRD)	~$293.00
Platform net profit (domestic card)	~$8.54
Platform net profit (international card)	~$2.34

⚠ 5% is Insufficient — Especially for International Cards

At 5%, the platform earns as little as **$1.71 per booking** when a customer pays with an international card and the piper is not GST registered. After accounting for infrastructure costs (Convex, hosting, email), support time, and the risk of occasional chargebacks, the platform would operate at a loss on a significant proportion of bookings.

SECTION 5

Platform Fee Recommendation

Recommendation

10%

Platform fee + GST
on each booking

This replaces the current 5% rate and provides:

- Positive net margin on both domestic and international cards
- Buffer for disputes, chargebacks, and refunds
- Below-market rate vs competitors (most charge 15–20%)
- Simple communication: "10% + GST service fee"

Comparison at $300 Booking — 5% vs 10% Platform Fee

Metric	5% (Current)	10% (Recommended)
Platform fee (incl. GST)	$17.25	$34.50
Platform GST to IRD	$2.25	$4.50
Stripe cost (domestic 1.7%)	~$5.66	~$5.80
Stripe cost (international 3.5%)	~$11.33	~$12.06
Net profit (domestic card)	**~$9.34**	**~$24.20**
Net profit (international card)	**~$3.67**	**~$17.94**
Customer pays (non-GST piper)	$315.00	$334.50
Customer pays (GST-reg piper)	$362.25	$379.50

Market Context

Platform Type	Typical Fee
Entertainment/talent marketplaces	15–25%
Event booking platforms	10–20%
Freelance service platforms (Fiverr, etc.)	20%
Highland Bagpiper (recommended)	**10% + GST**

At 10%: Estimated Annual Platform Revenue

Assuming 5 active pipers, each with 4 bookings per month at an average of $300: 20 bookings × $24.20 net ≈ **$484/month · $5,808/year** (domestic cards only). This grows linearly with piper adoption and is a sustainable foundation for the platform.

SECTION 6

Implementation Plan

The following steps transform the current system into a GST-aware, piper-friendly payment platform. Steps are ordered to avoid breaking changes — each step is safe to deploy independently.

Step 1 — Schema: Add GST Fields

1 **File:** `convex/schema.ts`

Add three optional fields to the `bagpipers` table: `gstRegistered` (boolean), `gstNumber` (string), `companyName` (string). Add a `feeBreakdown` object to the `bookings` table to store the full GST breakdown at time of payment. These are all optional — no migration needed, existing records are unaffected.

Step 2 — Fee Calculation Helper

2 **New file:** `convex/feeCalculations.ts`

A pure TypeScript module (no Convex runtime dependency) containing the `calculateFees()` function and the constants `PLATFORM_FEE_RATE = 0.10` and `GST_RATE = 0.15`. This is the single source of truth for all fee logic, importable by both `stripe.ts` and `bookings.ts`.

Step 3 — Piper Mutations: GST Support

3 **File:** `convex/bagpipers.ts`

Add the three new GST fields to `createProfile` and `updateProfile` mutation args. Add a dedicated `updateGstDetails` mutation for the Payments tab save action — keeps GST management separate from general profile editing.

Step 4 — Stripe Checkout: GST-Aware Fees

4 **File:** `convex/stripe.ts`

Replace the hardcoded `baseFee × 1.05` logic with `calculateFees()`. Pass the full fee breakdown as Stripe session metadata (stringified) so it survives through to the webhook. Update `application_fee_amount` to use the GST-inclusive platform fee.

Step 5 — Webhook: Persist Fee Breakdown

5

Files: `convex/stripe.ts` · `convex/stripeHelpers.ts`

Update the `checkout.session.completed` webhook handler to extract the fee breakdown from session metadata and pass it to `markPaid`. Update `markPaid` to persist it on the booking document so pipers can always see the breakdown of any payment.

Step 6 — Bookings: GST-Aware Estimates

6

File: `convex/bookings.ts`

Update `createBooking` initial fee calculation to use `calculateFees()`. Update quote notification messages to correctly describe whether the quoted amount is GST-exclusive (for registered pipers) or flat (for unregistered pipers).

Step 7 — Piper Profile UI: GST Section

7

File: `src/components/BagpiperProfile.tsx`

Add a `GstDetailsSection` component to the Payments tab. Includes: a GST registered toggle, conditional GST number and company name fields, and a live fee breakdown preview card that shows exactly what the customer will pay and what the piper will receive — updates dynamically as they change their GST status.

Step 8 — Piper Dashboard: Booking Fee Display

8

File: Piper bookings view component

Add a collapsible fee breakdown card to each paid booking in the piper's dashboard. Shows: their base fee, GST component (if registered), platform fee, total customer paid, and their received amount. Gives pipers the information they need for GST record-keeping.

SECTION 7

IRD Compliance Notes

Platform Obligations (Matihiko Mahi Limited)

- The platform must charge GST (15%) on its service fee for every booking
- GST is calculated as 15/115 of the GST-inclusive platform fee per booking
- The platform must file regular GST returns with IRD and remit collected GST
- The `feeBreakdown.platformGst` field stored per booking provides an accurate per-booking GST record for returns
- Consider generating a monthly platform GST summary report from booking data

Piper Obligations

- **GST-registered pipers:** Must file and remit the GST component of each booking with IRD as part of their regular GST returns. The piper receives the GST-inclusive amount from Stripe; the GST portion is not theirs to keep.
- **Non-registered pipers:** No GST obligation. Their entire received amount is income.
- The platform should advise pipers in writing (in the UI and in confirmation emails) of their GST obligations — see the dashboard fee breakdown card in Step 8.
- The `gstNumber` field is stored for record-keeping but cannot be validated in real-time (IRD does not provide a public validation API). Include a note: "Please ensure your GST number is correct — you are responsible for your GST compliance."

Record-Keeping Recommendation

When a booking is marked as paid, send the piper an automated email with the full fee breakdown. This gives them a paper trail for their GST returns and reduces support queries. The breakdown should include: booking date, event type, piper base fee, GST on their fee, platform fee, and their received amount.

Disclaimer

This document provides a technical and financial analysis of the platform's fee structure and is not legal or tax advice. Matihiko Mahi Limited should consult a qualified NZ accountant or tax

adviser to confirm GST treatment, particularly for any edge cases involving non-NZ customers or pipers.

Prepared by Claude Code · Highland Bagpiper Platform · Matihiko Mahi Limited · 24 March 2026